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Artificial Intelligence Applications – ITAI 2372

AI Usecases in Healthcare

“The medicine and healthcare industry focuses on the prevention, diagnosis, treatment, and management of diseases, aiming to enhance overall health through services delivered by a variety of trained professionals” (Medicine and Health Care Industry | Health and Medicine | Research Starters | EBSCO Research, n.d.). The healthcare industry serves patients regardless of their age, gender, and background.

AI in healthcare

One of the main applications of AI in healthcare is image diagnosis. An example of this application would be the model developed by Google Health in collaboration with Imperial College London; they designed and trained a model on X-ray images of more than twenty-nine thousand women. The task of the model was to identify any sign of breast cancer at early stages of the disease, when treatment can be more successful. This model brings more accuracy to the field, since previous screening programs were affected by high rates of false positives and false negatives. To test the accuracy of the model, researchers confronted two specialists who had access to the patients' medical records, while the model only had access to the X-ray images. This research concluded that the model was as good as two specialists and superior to a single doctor at spotting cancer.

Another impressive and hopeful application is APEX. Cesar de la Fuente is a biotechnologist from A Coruña, Spain (where I am originally from). He is the leader of an interdisciplinary team that has been working on biomedical innovations at the University of Pennsylvania. This team has developed a model called APEX, a system capable of analyzing the genetic material of millions of organisms (even extinct organisms like mammoths) to identify molecules with antibiotic potential. De la Fuente states that traditional methods take around six years to develop new candidates. On the other hand, this system reduces these times by identifying thousands of candidates in just a few hours (Magubane, 2024). This application is important because there has been a rise in drug-resistant bacteria in the last decade. This can lead to a health crisis since we can run out of antibiotics.

Job Displacement

AI has positive impacts on the healthcare industry, helping to solve some of the current issues that this industry is facing:

- Shortage of healthcare professionals: Automatic diagnosis by AI systems or telemedicine for towns that do not have enough healthcare professionals.
- Rising healthcare costs: Designing specific medical plans based on the patient's medical history.
- Administrative burden: AI tools can automate tasks like data entry or appointment scheduling.

But the introduction of AI brings new issues to the healthcare field, such as job displacement. Certain roles like medical coding or basic diagnosis can be automated. This does not affect current health professionals, but it decreases the opportunities for new healthcare professionals without experience. To mitigate this negative impact, it is essential to seek collaboration between AI systems and professionals instead of replacement (*The Impact of AI on the Healthcare Workforce: Balancing Opportunities and Challenges*, n.d.).

Transparency/Explainability

Transparency or explainability in AI refers to the capability to understand and interpret the data inputs and decision-making of algorithms ("Beyond the Black Box: Transparency Strategies for Healthcare AI," 2026). This problem is really concerning in the healthcare field because, even if the AI system can be more accurate than humans, it cannot explain its medical decisions to patients. This affects the informed consent principle, since patients would receive life-changing diagnoses generated by an AI system where neither the system nor the doctor can explain why it would benefit the patient.

Future Trends

The implementation of AI in medicine is going to change the direction of the field toward a more personalized medicine. AI will analyze genetic, environmental, and lifestyle data to create tailored treatment plans as well as predictive and preventive healthcare plans. AI will accelerate processes such as diagnosis and drug discovery, enhancing operations like scheduling, and creating fully automated laboratories (The Future of AI in Healthcare: Big Shifts to Expect in the Next Decade, 2026).

In conclusion, the AI application that impressed me the most was the use of APEX to discover new antibiotics, particularly the idea of extracting antibiotic potential from the genetic material of extinct organisms like the mammoth. What makes it even more remarkable is the scale of the acceleration: reducing a six-year process to just a few hours. On a personal level, knowing that the scientist behind this breakthrough, Cesar de la Fuente, studied and grew up in A Coruña makes this feel more real. This is a project I would like to be part of.

Citations:

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